Deliverable D10 records the outcome of Steps 1-3 in narrative form. D10 documents how assurance boundaries arise in practice for the scoped scenario and why certain hazards impose fundamental constraints on satisfiable autonomy claims. In doing so, it makes explicit where the requirements derived in Steps 3.2–3.4 remain valid only under bounded conditions and where those bounds are shaped by physical, organisational, or informational features of railway operation.

The role of D10 is therefore twofold. First, it acts as an assurance boundary, explicitly stating the conditions under which nominal requirements cease to be defensible and must instead trigger degraded operation or withdrawal of autonomy. Second, it serves as a direct input to Step 4, by indicating where metric definition must focus on uncertainty, integrity, and bounded confidence rather than nominal performance values. The detailed justification for these limitations, including relevant operational practice and historical precedent.

Within the previous sections of this chapter, twelve hazards were identified for the Swalwell Junction–Newcastle corridor, with three explored in depth: H01 (SPAD), H08 (degraded braking under low adhesion) and H11 (uncertainty regarding authority). Neither the three hazards discussed here nor the twelve hazards recorded in the Hazard Traceability Matrix (D9) are to be interpreted as exhaustive. Rather, they represent a first pass appropriate to demonstrating Safe Autonomy of Complex Railway Environments within a Digital space (SACRED) as a worked industrial case study. In an operational deployment, both the hazard register and the traceability structure would be expanded and iterated with stakeholders, and the traceability matrix would capture fault states across the full Operational Domain Model (ODM) from both ecosystem and ego perspectives, including explicit inter-hazard dependencies.

This appendix exists to justify why the three hazards above are treated as limiting, what those limitations imply for autonomy-enabling technology, and which boundaries must be respected such that later chapters can define key metrics that will ultimately constrain the Safety of the Intended Functionality (SOTIF/Safe Operating Concept (SOC)-style) argument.

0.0.1 H01: Signals Passed At Danger (SPAD)

H01 (SPAD) is routed from the more general phenomenon of a signal being passed at red (SPAR). Academic work that explicitly characterises the translation from "passed a signal at red" (an observable event) to "SPAD" (an operational hazard classification) is limited. One of the few sources that makes this distinction legible is Appendix A of Olsson's thesis *Mastering skills in an unpredictable world: simulator training for train drivers*,[1] which recounts the "43 situations" Swedish drivers are expected to learn. In Olsson's framing, "signal passed at red" is treated as a fact-state, whereas "SPAD" is the operational realisation of a hazard: its meaning depends on context, authority, and procedure.

In the UK, a SPAD is generally defined as an event where a train passes a stop signal without authority to do so.[2] SPADs are significant safety incidents and are commonly treated as pre-

[1] Niklas Olsson. *Mastering Skills in an Unpredictable World: Simulator Training for Train Drivers*. Luleå: Luleå University of Technology, 2024. ISBN: 978-91-8048-647-7.

[2] **https://www.orr.gov.uk/guidance-compliance/rail/health-safety/infrastructure/signals-passed-danger**.

cursors to collisions. However, the relationship between "train passes a red aspect" and "rules violation" is not one-to-one. There exist tightly constrained procedures under which passing a signal at danger is deliberately permitted, and there exist physical conditions under which a red signal may be passed despite correct driver intent and procedure. Conversely, even with safeguards (e.g., Automatic Warning System (AWS) and Train Protection and Warning System (TPWS)) in place, SPADs still occur due to combinations of human factors, infrastructure design, visibility limitations, and environmental conditions which can culminate in unsafe outcomes.

As one Associated Society of Locomotive Engineers and Firemen (ASLEF) driver put it during the professional elicitation that informed:[3] *"Thinking you know better than your signalling officer is how you get in trouble."* The significance of this remark, for autonomy, is that it highlights a tension: the cab may sometimes possess local information that the signaller does not, but the operational ecosystem remains the authority boundary to which the cab is accountable. If an autonomy stack ever claims that it "knew better" than the signaller, the assurance burden shifts to how that claim is justified, recorded, and audited.

A useful illustration of SPAD as a systemic phenomenon is the 2019 near-collision in New South Wales, documented by the ATSB in report RO-2019-009.[4] While initially framed in human-error terms, the investigation identifies an interacting network of contributing factors: an atypical signal configuration, signal positioning that could reasonably be mistaken as applying to a different track, and control book diagrams that did not resolve the ambiguity. The driver misinterpreted the applicable aspect, but the misinterpretation was shaped by infrastructure design, documentation practices, and training limitations. The point for this thesis is not that "human error" is irrelevant, but that SPAD is rarely a binary failure event: it is often a swiss-cheese alignment of perception, interpretation, authority, and braking capability within a complex operational ecosystem. This is precisely why autonomy cannot treat SPAD as a simple boundary breach divorced from context.

0.0.1.1 Authorised versus unavoidable red-signal passes

Certain red-signal passes occur under explicit authority and controlled procedure. As outlined in UK Rule Book Module S5,[5] signallers may authorise a train to pass a signal at danger under defined degraded-mode conditions (e.g., signal failure, controlled shunting, emergency recovery). These movements are coupled to strict safety conditions: route verification, protection of crossings where relevant, speed restrictions, and structured communication. Importantly, an authorised pass is not inherently risk-free. The Redcar level crossing collision (Rail Accident Investigation Branch (RAIB) Report R052025)[6] shows how procedural gaps (e.g., failure to secure the crossing) can still yield harm even when authority is nominally present.

Later revisions of Module S5[7] include "extreme emergency" provisions in which a driver may pass a signal at danger without contact. In this thesis we refer to this theoretical cate-

[3] Josh Hunter, John McDermid, and Simon Burton. "Insights from Railway Professionals: Rethinking Railway assumptions regarding safety and autonomy." In: Apr. 2024.

[4] **https://www.atsb.gov.au/publications/investigation_reports/2019/rair/ro-2019-009**.

[5] **https://www.rssb.co.uk/standards-catalogue/CatalogueItem/GERT8000-S5-Iss-1**.

[6] **https://assets.publishing.service.gov.uk/media/67ed3c5ce9c76fa33048c684/R052025_250403_Redcar.pd**

[7] **https://www.rssb.co.uk/standards-catalogue/CatalogueItem/GERT8000-S5-Iss-2**.

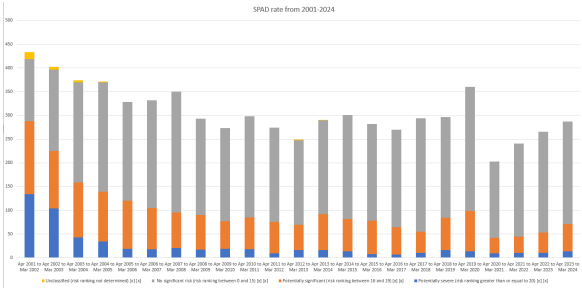

Figure 1: Signal Passed At Danger classifications in the UK between 2000–2025.

gory as an Unauthorised-Authorised Signal Passed At Danger (UASPAD), to emphasise the distinction between acting without contemporaneous permission or acting within a procedural exception that is itself authorised by the rule set. At present, publicly documented instances of UASPAD are difficult to identify, and informal driver commentary suggests that the scenario is more hypothetical than routine practice.[8]

Unavoidable red-signal passes refer to instances where a train passes a red signal despite correct intent and appropriate action, typically due to physical limitations such as extreme low adhesion or braking impairment. The Stonegate incident[9] exemplifies this: wet leaves and insufficient sand compromised braking such that the SPAD became effectively inevitable despite appropriate driver response.

0.0.1.2 Risk-ranked reporting versus operational meaning

Operational documentation tends to discuss SPADs in the authority/procedure framing above, but official reporting also classifies SPADs by risk ranking (Figure 1). Risk ranking methodology is described in the Office of Road and Rail (ORR) quality and methodology report[10] and in ORR rail safety statistics documentation.[11] However, the underlying quantitative details that separate, for example, "no significant risk" from "potentially severe" are not always transparent from public summaries. This matters for autonomy because it limits how directly we can map a proposed runtime metric to the regulator's published categories without additional access to the precise scoring model or its current revision.

H01 limitation: what this implies for technology and metrics

Limitation (assurance boundary). An autonomy claim cannot treat "red aspect passed" as a single failure condition. Assurance must be bounded by the system's ability to a. correctly identify and interpret the controlling signal in time, b. maintain sufficient braking capability to comply when required, and c. correctly resolve and evidence movement authority

[8] Within informal communication with drivers, UASPAD appear to be treated as an extreme hypothetical rather than a common operational pattern.

[9] **https://assets.publishing.service.gov.uk/media/5a7f5377e5274a2e8ab4b7ed/111117_R182011_Stonegate**

[10] Statistics Office of Road and Rail. *Rail Safety - 2024*. Tech. rep. ORR - RIDDOR, Sept. 2024. URL: `https://dataportal.orr.gov.uk/statistics/health-and-safety/rail-safety/`.

[11] **https://dataportal.orr.gov.uk/statistics/health-and-safety/rail-safety/**.

(including degraded-mode procedure where applicable). If authority is unresolved or braking margin is insufficient, continued movement cannot be assured.

Technology implication. The system must be able to perceive and interpret signals and their applicability (identity/track association), and it must be able to communicate with signalling (or enact a pre-defined emergency procedure) such that an authorised pass can be requested, recorded, and audited. It must also be able to justify a SPAR-to-SPAD classification in a way that is compatible with investigation practice.

Metric consequence for Step 4/SOC. Step 4 must define a signal-interpretation integrity measure (including applicability confidence), a braking margin / stopping-distance bound, and an authority-resolution state (resolved/unresolved, with deadlines). The SOC must gate permissive operation on these conditions.

0.0.2 H08: Low adhesion and degraded braking capability

Hazard H08 (low adhesion) is grounded in well-known operational failure modes. The Salisbury Tunnel Collision (RAIB Report R12/2023)[12] involved a train passing a stop signal and colliding with another train; the investigation identified extremely low adhesion due to leaf contamination as a key causal contributor. While the immediate aftermath of such events is often framed around driver action, investigations regularly show that correct driver intent can be defeated by physical limits of wheel–rail adhesion and braking performance.

Network Rail (NR)'s Weather Advisory Task Force (WATF) report[13] explicitly identifies weather as a safety-critical influence on railway operation and emphasises that weather effects relevant to adhesion cannot be treated as static planning assumptions. Adhesion-relevant conditions can diverge rapidly from forecast, vary over short spatial scales, and therefore demand operational monitoring and response rather than reliance on pre-computed environmental models.

For this reason, it is not useful to treat "weather" as an ODM axis to be exhaustively enumerated. Attempting to include all permutations of weather as domain partitions would produce an unbounded and continuously shifting representation. Following the WATF framing, weather is positioned as an ecosystem-level responsibility: the wider railway system is assumed to monitor, interpret, and communicate weather-related effects as they relate to adhesion and braking capability.

The autonomy-relevant limitation is therefore not that the system must "model weather", but that it must be provided with (or must infer) operationally meaningful adhesion/braking information. As captured in D9, this becomes a requirement that adhesion-relevant state is available within the operational domain at runtime; without it, braking performance cannot be bounded.

[12] UK RAIB. *Collision between passenger trains at Salisbury Tunnel Junction.* Rail Accident Investigation: Interim Report Report IR1/2022. Feb. 2022. (Visited on 04/27/2023).

[13] **dame_weather_2021**.

H08 limitation: what this implies for technology and metrics

Limitation (assurance boundary). Assurance of speed compliance and stop compliance is only claimable within the adhesion bounds assumed by the braking model. If adhesion state is unknown, stale, or outside the bounded range, assurance cannot be claimed for nominal braking performance.

Technology implication. Either the ecosystem must provide actionable adhesion/braking information to the cab (with appropriate timeliness and coverage), or the cab must infer a conservative braking capability from onboard sensing and operational feedback. In either case, the cab must degrade behaviour when adhesion integrity is not established.

Metric consequence for Step 4/SOC. Step 4 must define an "adhesion/braking capability" metric (continuous bound or conservative discrete classes) plus an integrity/timeliness condition. The SOC must gate relative braking, approach speeds, and degraded-mode policies on these bounds.

0.0.3 H11: Uncertainty regarding authority and communication

H11 concerns uncertainty or conflict in movement authority: cases where the cab cannot clearly determine whether it is permitted to proceed, or where different authority-relevant signals (aspect, protection system indications, instructions, local knowledge) do not align. Unlike H01/H08, there is limited historical precedent for formalising this as its own hazard because, in conventional operation, authority resolution is ultimately mediated by human procedure and *post-hoc* investigation: if a SPAR becomes a SPAD, the driver's actions are examined and judged "appropriate" or not. That raises a core question for autonomy: what constitutes "appropriate" when the decision-maker is a system, and what evidence must it produce?

If publicly documented instances of UASPAD are rare or absent, one interpretation is that although the cab is theoretically the final actor, in practice it is expected to treat signaller authority as dominant except in extreme and procedurally framed emergencies. The signalling handbook[14] places emphasis on gathering sufficient information to establish exactly where movement authority lies when doubt exists. For autonomy, that implies two requirements: the capability to communicate with signalling and the capability to represent and reason about authority as a state that can be unresolved.

A further practical complication is that signaller–driver communications are not purely formal: shorthand, local dialect, and region-specific phrasing may dominate in routine talk. Informal correspondence with signalling staff suggests that "communication" is not a trivial interface to automate. This pushes H11 beyond a simple "add a radio link" requirement (and perhaps add speech generation and interpretation); it becomes a question of whether the autonomy concept relies on free-form language understanding, or whether it can constrain communications into structured, auditable protocol.

[14] **https://www.rssb.co.uk/standards-catalogue/CatalogueItem/GERT8000-S5-Iss-2**.

H11 limitation: what this implies for technology and metrics

Limitation (assurance boundary). Autonomous operation is only claimable when movement authority can be resolved within a bounded time/distance window. If authority is unresolved, conflicting, or communication integrity is insufficient to resolve it, the autonomy claim must degrade to a minimum-risk policy and escalation.

Technology implication. If the concept relies on conversational interaction, the assurance burden becomes extremely high (coverage of phraseology, ambiguity handling, auditability). A more defensible route is to constrain authority exchange to structured, standardised messages (even if delivered via human speech), with explicit confirmation and logging.

Metric consequence for Step 4/SOC. Step 4 must define an authority-resolution metric (resolved/unresolved plus deadlines), and a communication integrity condition sufficient for authority establishment. The SOC must gate permissive movement on those conditions.